Junseo Kim

PhD Student · Robotics Engineer

Stockholm, Sweden

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Summary

A robotics researcher object interaction using dexterous hand, and humanoid. Studying algorithms that are robust generalizable, better sim2real transfer, with the goal of achieving human-level dexterity for robot manipulation. Currently, a PhD Student at the KTH royal institute of Technology under supervision of Prof. Danica Kragic. Previously worked as a visiting research scholar at the BIG Lab at CMU, Pittsburgh, PA, United States.

Field of Interest

My research interests include, but are not confined to:

- Dexterous Manipulation
- Humanoid whole-body control
- Motion Retargeting from MoCap, Video data
- Wireless signal processing for Robotics application
- Visual-Inertial Odometry

Education

* includes 18-months military service

KTH Royal Institute of Technology — *Stockholm, Sweden*

Ph.D. in Computer Science — *Aug. 2026 - Present*

- Advised by Prof. Danica Kragic

Delft University of Technology (TU Delft) — *Delft, Netherlands*

M.S. in Robotics — *Sept. 2024 - Jul. 2026*

- Advised by Prof. Javier Alonso-Mora

Toronto Metropolitan University — *Toronto, ON, Canada*

B.Eng. in Mechanical Engineering — *Sept. 2018 - Jun. 2024**

- Advised by Prof. Sajad Saeedi

Work Experience (Including Internship)

Visiting Research Scholar — *Pittsburgh, PA, United States*

Bot Intelligence Group, CMU — *Aug. 2025 - Jun. 2026*

- Advised by Prof. Jean Oh.
- Conducted research on video action model using kinematic rendering.

Publications

Underline denotes co-first authors or co-corresponding authors

International Journal

1. **Junseo Kim**, Matthew Lisondra, Yeganeh Bahoo, and Sajad Saeedi, "Inverse k-visibility for RSSI-based Indoor geometric mapping", *Autonomous Robots, Springer Nature*, vol. 50, no. 2, pp. 17, 2026, `https://doi.org/10.1007/s10514-026-10243-w`
2. Ishaan Mehta, **Junseo Kim**, Sharareh Taghipour, and Sajad Saeedi, "M^{3}RS: Multi-robot, Multi-objective, and Multi-mode Routing and Scheduling", *Journal of Field Robotics*, 0: 1–16, `https://doi.org/10.1002/rob.70288`
3. Matthew Lisondra, **Junseo Kim**, Glenn Takashi Shimoda, Kourosh Zareinia, and Sajad Saeedi, "TCB-VIO: Tightly-Coupled Focal-Plane Binary-Enhanced Visual Inertial Odometry", *IEEE Robotics and Automation Letters*, vol. 10, no. 12, pp. 12341-12348, Dec. 2025, `https://doi.org/10.1109/LRA.2025.3619774`

International Conference

1. **Junseo Kim**, Guido Dumont, Xinyu Gao, Gang Chen, Holger Caesar, and Javier Alonso-Mora, "MobileOcc: A Human-Aware Semantic Occupancy Dataset for Mobile Robots," in *European Conference on Computer Vision (ECCV)*, 2026. Accepted. To appear.
2. Pablo Ortega-Kral, Eliot Xing, Arthur Bucker, Vernon Luk, **Junseo Kim**, Owen Kwon, Angchen Xie, Nikhil Sobanbabu, Yifu Yuan, Megan Lee, Deepam Ameria, Bhaswanth Ayapilla, Jaycie Bussel, Guanya Shi, Jonathan Francis, and Jean Oh, "RIO: Flexible Real-Time Robot I/O for Cross-Embodiment Robot Learning," in *Robotics: Science and Systems (RSS)*, 2026. Accepted. To appear.
3. Huichan Seo, Sieun Choi, Minki Hong, Yi Zhou, **Junseo Kim**, Lukman Ismaila, Naome Etori, Mehul Agarwal, Zhixuan Liu, Jihie Kim, and Jean Oh, "Exposing Blindspots: Cultural Bias Evaluation in Generative Image Models," in *Proceedings of IASEAI Conference*, 2(1), 724–736, `https://ojs.aaai.org/index.php/IASEAI/article/view/43063`
4. **Junseo Kim**, Jill Aghyourli Zalat, Yeganeh Bahoo, and Sajad Saeedi, "Structure from WiFi (SfW): RSSI-Based Geometric Mapping of Indoor Environments," in *American Control Conference (ACC), Toronto, ON, Canada*, 2024, pp. 259-264, `https://doi.org/10.23919/ACC60939.2024.10644833`.
5. Matthew Lisondra, **Junseo Kim**, Riku Murai, Kourosh Zareinia, and Sajad Saeedi, "Visual Inertial Odometry using Focal Plane Binary Features (BIT-VIO)," in *IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan*, 2024, pp. 1661-1668, `https://doi.org/10.1109/ICRA57147.2024.10610838`

Honors & Awards

External Awards

2023	**Korean Canadian Scientists Scholarship Foundation Award**, AKCSE	*Toronto*

Internal Awards

2025	**Justus and Louise van Effen Research Grant**, TU Delft	*Delft*
2024	**Justus and Louise van Effen Excellence Scholarship**, TU Delft	*Delft*
2024	**Dean's List**, TMU	*Toronto*
2023	**UIRO(Undergraduate Interdisciplinary Research Opportunities) Award**, TMU	*Toronto*

Academic Services

2026	**Reviewer**, ICRA, ECCV
2025	**Reviewer**, IROS